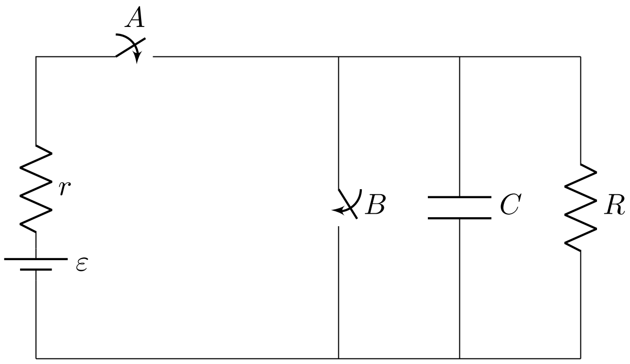
# Physics 5C Discussion (Week 7)

Department of Physics and Astronomy, UCLA

**Problem 1**: Jinyan in the lab has designed another circuit, with a battery that supplies emf $\varepsilon$ and internal resistance $r$. There are switches $A$ and $B$, capacitor $C$, and resistor $R$ in the circuit.




At the initial time, the parallel-plate capacitor **is already fully charged**.

**Part (a)** The battery has internal resistance $r$. If both of the switches are closed, find symbolically the current going through the battery (the leftmost path), and find the terminal voltage of the battery.

**Part (b)** What switch(es) should Jinyan close to keep the capacitor fully charged?

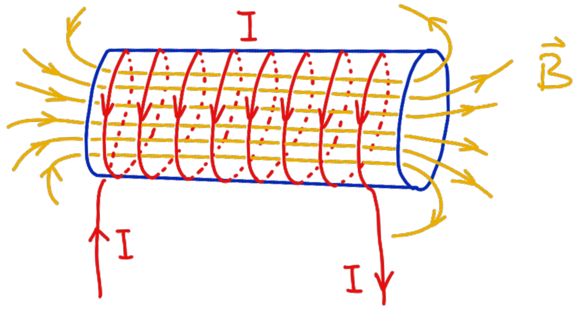
**Part (c)** While keeping the capacitor fully charged, Jinyan inserts a dielectric filling between the two plates of the parallel-plate capacitor. Find the change in voltage across the capacitor, and the change in energy stored in the capacitor, in terms of dielectric constant $\kappa$.

**Part (d)** If all switches are open at the initial time, what will happen to the capacitor?

**Part (e)** What will be the fastest way to discharge the capacitor, and why?

**Problem 2**: Current produces a magnetic field.

If we wind the wire into a solenoid, the magnetic field inside the solenoid produced by the current is uniform,



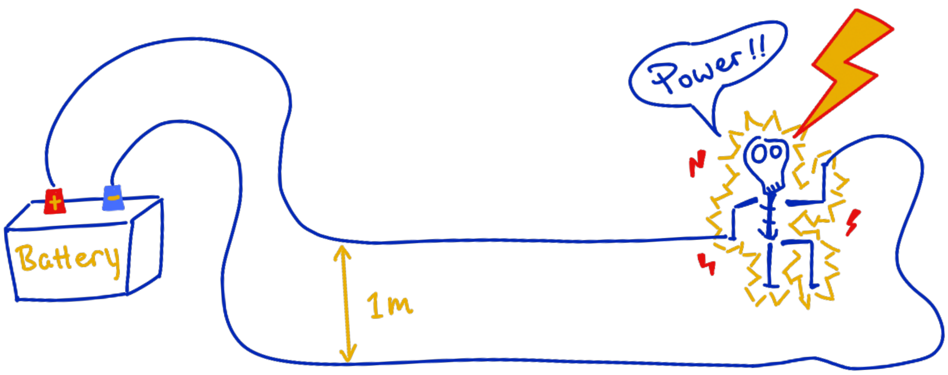

$$|\vec{B}| = \mu_0 N I / L\,,$$

where $\mu_0$ is a constant, $N$ is the number of "wire loops," $I$ is the current in the wire, and $L$ is the solenoid length. The direction of $\vec{B}$ is determined by the right-hand rule.

**Part (a)** Explain why the magnetic field inside a long (but has a finite length) solenoid is stronger and more uniform than the field outside.

**Part (b)** Jinyan needs a 1 T magnetic field for his science innovation. He wants to build it using a gigantic solenoid. He has a wire that is in total 100 m long, and he plans to wind the wire into a solenoid that has radius $r = 1$ m and a length of 10 m. How much current does he need in the wire to produce the amount of magnetic field that he wants.

**Problem 3**: Jinyan is doing some weird experiment again: He builds a circuit and places two straight wires close to each other as shown. Then he sees that the two wires repel each other. Please educate him on how to write his lab report.




**Part (a)** In what direction are the electrons in the wires moving?

**Part (b)** Why do the two wires repel each other?

**Part (c)** If the current in the wire is 50 $\mu$A, the distance between the two wires is 1 m, and an electron in the top wire is moving 1 nm/s, what is the repelling force experienced by the electron? Include both direction and magnitude in your answer.

## Problem 1

The key idea in this problem is to think about what the switches do to the circuit. Switch $A$ decides whether the battery is connected to the rest of the circuit. Switch $B$ provides a direct wire connecting the top and bottom nodes on the right. Since the capacitor is initially fully charged, it behaves like an open circuit if we leave it alone, but it can discharge if we give it a path.

**Part (a)** If both switches are closed, then switch $B$ acts like an ideal wire directly connecting the top and bottom nodes on the right side. That means the resistor $R$ and the capacitor are both shorted out, so the external part of the circuit has zero resistance. Therefore the battery is effectively short-circuited, and the only resistance left in the entire loop is the battery's internal resistance $r$. Thus the current through the battery is

$$I_{\text{battery}} = \frac{\varepsilon}{r}\,.$$

The terminal voltage of a battery is

$$V_{\text{terminal}} = \varepsilon - I_{\text{battery}}r = \varepsilon - \left(\frac{\varepsilon}{r}\right) r = 0\,.$$

Here we see that the battery current is $\varepsilon/r$, and the terminal voltage is zero.

** There is one more physical thing happening here: once switch $B$ is closed, the already-charged capacitor will also discharge through that short branch. In an ideal circuit model that discharge is extremely fast, but it does not change the steady battery current we just found.

**Part (b)** To keep the capacitor fully charged using the battery, we should close switch $A$ and leave switch $B$ open. Why? If $A$ is open, the battery is disconnected and cannot maintain the capacitor's charge. The capacitor-resistor loop on the right-hand side of the circuit allows the capacitor to discharge. If $B$ is closed, the capacitor is shorted and will discharge. So the correct choice is to close $A$ only and keep $B$ open.

In that configuration, as long as the capacitor is fully charged, we can view it as an open switch in the circuit. Then the only steady current in the external circuit is through the resistor $R$ (which can be viewed as in series with the little resistor $r$ connected to the battery). The battery current is then

$$I_{\text{battery}} = \frac{\varepsilon}{R+r}\,,$$

and the battery terminal voltage is

$$V_{\text{terminal}} = \varepsilon - I_{\text{battery}}r = \varepsilon - \frac{\varepsilon r}{R+r} = \frac{\varepsilon R}{R+r}\,.$$

This is also the voltage across the fully charged capacitor in part (c).

**Part (c)** While keeping the capacitor fully charged, we keep the same switch configuration as in part (b): $A$ closed and $B$ open. Therefore, the capacitor remains connected across the battery terminals, so its voltage is fixed by the battery terminal voltage,

$$V_C = \frac{\varepsilon R}{R + r}\,.$$

Now I inserts a dielectric with dielectric constant $\kappa$. The capacitance for sure will increase from $C$ to

$$C' = \kappa C\,.$$

Since the capacitor remains connected to the battery, the voltage across it does not change, and more charges are pumped to the capacitor by the battery. Thus $V_C$ stays the same but $Q$ increases. The initial stored energy is

$$U_i = \frac{1}{2} C V_C^2\,,$$

while the final stored energy is

$$U_f = \frac{1}{2}(\kappa C) V_C^2 = \kappa U_i\,.$$

Therefore the change in stored energy is

$$\Delta U = U_f - U_i = \frac{1}{2}(\kappa - 1) C V_C^2 = \frac{(\kappa - 1)C}{2}\left(\frac{\varepsilon R}{R + r}\right)^2\,.$$

So the voltage stays the same, while the stored energy increases by a factor of $\kappa$.

**Part (d)** If all switches are open initially, then the battery is disconnected from the top node, but the capacitor is still connected in parallel with the resistor $R$. Therefore the charged capacitor now has a closed path through the resistor and will discharge through the resistor.

As time goes on, the capacitor's charge, voltage, and stored energy all decrease to zero, and the stored energy is dissipated as heat in the resistor. In the standard RC language, the discharge is exponential,

$$Q(t) = Q_0 e^{-t/(RC)}\,, \qquad V_C(t) = V_0 e^{-t/(RC)}\,,$$

where $Q_0$ is the amount of charge stored on one of the plates of the parallel-plate capacitor when the capacitor is fully charged, and $V_0$ is the corresponding voltage difference across the two plates of the capacitor. In conclusion, in our case when two switches are open, the capacitor will not stay charged forever; it will gradually discharge through $R$.

**Part (e)** The fastest way to discharge the capacitor is to close switch $B$ while keeping switch $A$ open. The reason is that switch $B$ places an ideal wire directly across the capacitor, so the capacitor sees the smallest possible resistance. In the idealized circuit model, the effective discharge resistance is zero, so the discharge time constant is

$$\tau = R_{\text{eff}} C = 0\,.$$

That means the capacitor discharges essentially immediately in the ideal model. We keep $A$ open so that we do not also short-circuit the battery.

If both switches are open, the capacitor discharges through $R$. As we argued in part (d), the discharge time constant is then $RC$, which is way longer if $R$ is significant.

## Problem 2

**Part (a)** A solenoid is just many circular current loops stacked next to one another. Each loop produces a magnetic field. Inside the solenoid, the magnetic fields from neighboring loops point mostly in the same direction, so they add together. In the picture, based on the direction of current and using your right-hand rule, you can check that each loop produces a magnetic field pointing to the right at its center. The fields produced by these loops add together, and thus the field inside the solenoid is pointing to the right. Outside the solenoid, the opposing magnetic field contributions from the top and bottom wire segments of each loop mostly cancel (similar to what happens to electric field outside a finite-size parallel-plate capacitor). Thus the magnetic field inside the solenoid is much stronger than outside.

One needs some calculus trick to show why the field inside is uniform, but we can examine from the geometry that the field inside the solenoid, towards the middle of the solenoid is also more uniform than outside, and than the left and right ends of the solenoid. This is because far from the two ends towards the middle of the solenoid, each turn of the solenoid contributes in almost the same way, and the turns are closely stack together. While at the two ends, the symmetry is less perfect, so the field becomes less uniform there.

**Part (b)** The given formula is

$$|\vec{B}| = \mu_0 N I / L \,.$$

We are told that the total wire length is 100 m and the solenoid radius is $r = 1\,\mathrm{m}$. Each loop uses circumference

$$2\pi r = 2\pi(1\,\mathrm{m}) = 2\pi\,\mathrm{m}\,,$$

so the number of turns in the solenoid is

$$N = \frac{100}{2\pi} \approx 15.9\,.$$

Now solve the magnetic-field formula for the current. Rearranging the given formula,

$$I = \frac{BL}{\mu_0 N}\,.$$

We can plug in the numbers to get

$$I = \frac{(1\,\mathrm{T})\,(10\,\mathrm{m})}{\mu_0(50/\pi)} = \frac{10\pi}{50\,(4\pi\times 10^{-7})}\,\mathrm{A} = 5.0\times 10^{5}\,\mathrm{A}\,.$$

Note that the solenoid length $L$ is not the total length of the wire.

## Problem 3

The figure shows one long circuit loop whose top and bottom segments run parallel to each other. Current comes out of the positive terminal of the battery, travels through the top wire to the right, then goes down through the load on the right, and finally returns through the bottom wire to the negative terminal. Therefore the two long parallel wires carry currents in antiparallel directions.

**Part (a)** The conventional current in the top wire runs to the right, and the conventional current in the bottom wire runs to the left. Equivalently, because electrons move opposite to the conventional current, the electrons in the top wire drift to the left, while the electrons in the bottom wire drift to the right.

**Part (b)** Two parallel wires carrying current exert magnetic forces on each other. The standard rule is same-direction currents attract, opposite-direction currents repel. In this figure the currents in the two long straight wires are opposite, so the magnetic force between the wires is repulsive. But here is the "origin" of the rule, using the magnetic field and the Lorentz force law. The bottom wire creates a magnetic field at the location of the top wire, with magnitude given by

$$|\vec{B}| = \frac{\mu_0 I}{2\pi r}$$

and direction determined by the right-hand rule. In this case, the direction of the magnetic field created by the bottom wire (where current is leftwards) at the top wire location is into the page. The moving charges in the top wire (positive charge to the right, or electrons drifted to the left) feel a Lorentz force

$$\vec{F} = q\vec{v} \times \vec{B}\,.$$

That force points away from the bottom wire as you can verify using the right-hand rule again. By the same argument, the bottom wire is pushed away from the top wire, so the two wires repel each other.

**Part (c)** The current in the wire is $I = 50\,\mu\mathrm{A} = 5 \times 10^{-5}\,\mathrm{A}$, the wire separation is $d = 1\,\mathrm{m}$, and the electron speed is $v = 1\,\mathrm{nm/s} = 10^{-9}\,\mathrm{m/s}$. Then the magnetic field produced by the bottom wire at the location of the top wire is

$$B = \frac{\mu_0 I}{2\pi d} = \frac{(4\pi \times 10^{-7})(5 \times 10^{-5})}{2\pi(1)}\,\mathrm{T} = 1.0 \times 10^{-11}\,\mathrm{T}\,.$$

The magnetic force magnitude on the electron is

$$F = |q|vB = (1.6 \times 10^{-19})(10^{-9})(1.0 \times 10^{-11}) = 1.6 \times 10^{-39}\,\mathrm{N}\,.$$

The direction of this force is upward, meaning away from the bottom wire.


Madison, the capital city of Wisconsin, one of my favorite cities in the United States.
Pictured in March 2024, the year I applied for my Ph.D. program.